

Panos Oikonomou

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Education

New York University (NYU)

New York US | 2023 - now

PhD student in High Energy theory, Center for Cosmology and Particle Physics | **JAGA Fellow**

New York University Abu Dhabi (NYUAD)

Abu Dhabi, UAE | 2018 - 2022

Bachelor of Science. **Double Major in Physics and Mathematics**, Engineering Minor | **GPA: 3.87**

Research

Center for Cosmology and Particle Physics (CCPP), NYU

New York City, US | 2023 - now

Research Assistant, Particle Physics (full-time)

- **Charting Impurities at Tricriticality**
 - Classifying symmetry preserving conformal defects in the tricritical Ising minimal model via generalized gauging, with applications to boundary RG flows.
 - Analyzing invariance under topological line operators to constrain allowed defect RG endpoints.
- **Bootstrapping Generalized Symmetries**
 - Deriving symmetry-induced constraint inequalities in the numerical conformal bootstrap using categorical symmetries.
 - Implementing numerically using SDPB.
- **Dualities in Floquet Codes**
 - Exploring electric–magnetic duality in 2D lattice models with fusion-category topological order.

Center for Astro, Particle, and Planetary Physics (CAP³), NYUAD

Abu Dhabi, UAE | 2022 - 2024

Research Assistant, Particle Physics (full-time)

- **Quantization of Dissipative Systems in Hydrodynamics**
 - Effective Field Theory of phonons in viscous liquids.
- **Dark Matter Freezout During Reheating**
 - Developing a numerical method to solve the Boltzmann equation for arbitrary source terms to predict dark matter abundance in a model-independent way under different thermodynamic initial conditions.
- **Light-I Satellite Terrestrial Gamma-Ray Flash search.**
 - Led analysis of the LIGHT-I satellite terrestrial gamma-ray flash search; developed Python/C++ analysis pipeline and background modeling.

Center for Quantum and Topological Systems (CQTS), NYUAD

Abu Dhabi, UAE | 2022 - 2023

Research Assistant, Mathematical Physics

- **Quantum Information using Homotopic Type Theory**
 - Developed a framework for expressing Hilbert space symmetries using Homotopic Type Theory.
 - Built a Haskell library interpreting quantum circuits as braids.

Astroparticle Laboratory, NYUAD

Abu Dhabi, UAE | 2019 - 2022

Undergraduate Research Assistant, Astroparticle Physics (part-time)

- **Acoustic Signals in Viscous Liquids**
 - Modeled acoustic signals from charged particles in viscous media using hydrodynamics and QFT, applied to WIMP searches at XENONnT.
 - Derived spectral solutions to strongly damped wave equations, formulated an effective field theory for dissipative quantization, and developed a parallel AMR finite-element C++ framework (first-author publication).
- **Multilayered Dielectric Haloscope Investigation (MuDHI)**
 - Wrote a background model simulation in GEANT4, for this tabletop dark matter experiment.
 - Analytically predicted the optimal experimental setup. Co-authored paper.

Publications - <https://orcid.org/0000-0003-2296-6415>

* *Corresponding author*, **Principal Investigator*

- Manenti L.*; Pepe C.; Sarnof I.f; Ibrayev T.; **Oikonomou P.**; Knyazev A. *Dark counts in optical superconducting transition-edge sensors for rare-event searches* (2024)
- The XENON Collaboration. *A next-generation liquid xenon observatory for dark matter and neutrino physics* (2023)
- **Oikonomou, P.***; Sarnoff I.; Manenti L.; Arneodo, F.*; *Acoustic detection potential of single minimum ionizing particles in viscous liquids*, Physical Review Research (2023)
- DiGiovanni, A.*; AlKindi, L.; **Oikonomou, P.**; ..., Arneodo, F.*; *Detecting Terrestrial Gamma-ray Flashes with Cubesats: the LIGHT-I Payload instrument*; JINST (2023)
- Kalos, S.*; Arneodo, F.*; ... **Oikonomou, P.**; ... Els S.* (2022) *Absolute Photoelectric Yield Measurements of the Rashid Lunar Rover Materials at the ELETTRA Synchrotron Facility*, Journal of Geophysical Research. (2024)
- DiGiovanni, A.*; Arneodo, F.; ... **Oikonomou, P.**; ... (2022) *The Scientific Payload of LIGHT-I: A 3U Cubesat Mission for the detection of Terrestrial Gamma-Ray Flashes*, Nuclear Inst. and Methods in Physics Research A.
- Manenti, L.*; Mishra, U.*; ... **Oikonomou, P.**; ... Arneodo, F.* (2021) *Search for dark photons using a multilayer dielectric haloscope equipped with a single-photon avalanche diode*, Physical Review D.
- **Oikonomou, P.***; Pappas, S. (2020) *Decentralized Bioinspired Non-Discrete Model for Autonomous Swarm Aggregation Dynamics*, Applied Science.

Teaching

Quantum Mechanics II, Physics, NYU	New York, US 2026
Computational Physics, Physics, NYU	New York, US 2025
General Physics II, Physics, Summer, NYU	New York, US 2025
Statistical Physics, Physics, NYU	New York, US 2025
Quantum Mechanics I, Physics, NYU	New York, US 2024
Fluids, Mathematics, NYU	New York, US 2024
Multivariable Calculus, Mathematics, NYUAD	New York, US 2022 - 2023
Conserving our Global Heritage Through Science, Core Curriculum, NYUAD	Abu Dhabi, UAE 2022
Computer Programming for Engineers, Engineering, NYUAD	Abu Dhabi, UAE 2019 - 2021

Seminars

European Organization for Nuclear Research (CERN)	Geneva, Switzerland 2022
CERN Winter School on Supergravity, Strings, and Gauge Theory 2022	
National Technical University of Athens (NTUA)	Athens, Greece 2018
Data Structures and Algorithms for International Olympiad of Informatics	

Awards

James Arthur Graduate Associate Fellowship (CCPP)	New York, US 2025-2026
Awarded by the Center for Cosmology and Particle Physics for research potential.	
NYU Outstanding Teaching Award (NYU)	New York, US 2025
Awarded by the College of Arts and Science at NYU for teaching.	